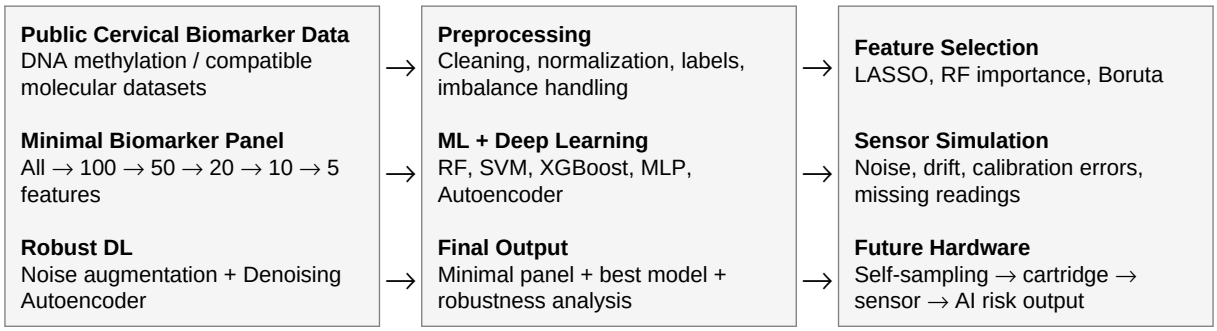
AI & Deep Learning-Based Early Cervical Precancer Risk Screening

Proposed Research Goal

Develop an AI/DL framework that identifies a small set of molecular biomarkers associated with cervical precancer or high-risk abnormal patterns and evaluates whether predictions remain reliable under realistic low-cost sensor measurement errors. The project does not claim clinical cancer diagnosis.

Proposed Research Flow



What We Will Do (3 Months)

Month 1	Prepare public datasets; preprocessing; baseline ML; identify candidate biomarkers.
Month 2	Feature reduction experiments; compare ML vs MLP/autoencoder; determine minimal biomarker panel.
Month 3	Simulate sensor noise/drift/missing values; develop robust models; evaluate denoising autoencoder.

Key Novelty

1. **Minimal biomarker panel:** study how performance changes as thousands of molecular features are reduced to a small panel suitable for future sensing. 2. **Sensor-aware AI:** evaluate models under simulated measurement noise, drift and missing readings rather than only clean laboratory data. 3. **Robust Deep Learning:** compare normal models, noise-augmented models and denoising autoencoders.

Expected Output

A candidate minimal biomarker panel, ML/DL performance comparison, performance-vs-number-of-biomarkers analysis, sensor robustness curves, and a robust AI framework for future biosensor integration.

Future Hardware Concept

Self-collected cervicovaginal swab → **sample preparation** → **disposable cartridge** → **detection of selected biomarkers** → **electrical/optical sensor signals** → **AI model** → **Low-risk / High-risk abnormal biomarker pattern.**
The current project develops the computational foundation; physical sensor development and clinical testing are future work.

Feasibility & Limitations

Feasible now: public-data analysis, ML/DL, feature selection and sensor-error simulation. **Limitations:** no patient samples or clinical access; public datasets may be small and heterogeneous; simulated sensor noise is not equivalent to real hardware; selected biomarkers require later biological and sensor feasibility validation. Therefore, the system will be presented as a computational research framework, not a clinically validated cancer diagnostic kit.